

# MATH-273: Statistics and Probability Theory

## Assignment on 05422173

2<sup>nd</sup> Year 1<sup>st</sup> Semester 2023

Dept. of CSE, J.K.K.N.I.U.

Q1) The following data refer to the ages of 60 employees of a firm.

|    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| 33 | 41 | 21 | 25 | 36 | 38 | 35 | 36 | 35 | 37 | 42 | 30 | 35 | 37 | 36 |
| 38 | 30 | 54 | 40 | 48 | 15 | 28 | 51 | 42 | 25 | 41 | 30 | 27 | 42 | 36 |
| 28 | 26 | 37 | 54 | 44 | 31 | 36 | 40 | 36 | 22 | 30 | 31 | 19 | 48 | 16 |
| 42 | 32 | 21 | 22 | 40 | 43 | 42 | 39 | 38 | 37 | 33 | 49 | 47 | 46 | 48 |

- Construct an exclusive frequency distribution with suitable class interval.
- Draw histogram, frequency polygon and Ogive curve with the frequency distribution.
- Display the data set by a stem-and-leaf plot.

**Solution:**

- Range:  $R = 54 - 15 = 39$

Since the number of classes can be 5 to 25 so the class interval is chosen within  $\frac{R}{25}$  and  $\frac{R}{5}$ .

$$\therefore \frac{R}{25} = \frac{39}{25} = 1.56 \text{ and } \frac{R}{5} = \frac{39}{5} = 7.8$$

So, conveniently class interval is chosen 5.

| Class interval (Ages) | Frequency | Min-value |
|-----------------------|-----------|-----------|
| 15-20                 | 3         | 17.5      |
| 20-25                 | 4         | 22.5      |
| 25-30                 | 6         | 27.5      |
| 30-35                 | 9         | 32.5      |
| 35-40                 | 17        | 37.5      |
| 40-45                 | 12        | 42.5      |
| 45-50                 | 6         | 47.5      |
| 50-55                 | 3         | 52.5      |

- Histogram:** Class intervals on X axis and corresponding frequencies on Y axis.

**Frequency Polygon:** Mid values on X axis and corresponding frequencies on Y axis then connect the points.

**Ogive:** Upper class limits on X axis and corresponding cumulative frequencies on Y axis.

- First digits are stem and last digits are leaves.

Q2) The owner of a dress house wants to know the weekly requirements of different sizes of shirts demanded by customers. For this, he collected the information of different shirt sizes by the dress house in last week. The data are given below

M XL M L M S XL M L L XL L M S XL  
M M XL S M M M M L S M S S S M

S S M M M S M M M L M L L M XL  
L XL L M S S L M M L XL S S XL M

Here, S, M, L, XL indicate small, medium, large and extra-large sizes of shirts respectively.

- Summarize the data in a frequency distribution and mention which shirt size is most frequent.
- Draw a bar diagram for the frequency distribution and pie chart for the percentage distribution.
- If the owner wants to store 100 shirts for the next week, what plan should he take?

[Hint: iv) Percent of each size will be found in (ii). Owner will store the sizes according to these percent]

**Q3)** The following table gives the number of workers in different categories of four garments factory.

| Worker | Factory-A | Factory-B | Factory-C | Factory-D |
|--------|-----------|-----------|-----------|-----------|
| Male   | 100       | 250       | 300       | 250       |
| Female | 600       | 750       | 850       | 900       |
| Girl   | 250       | 150       | 250       | 250       |
| Boy    | 50        | 50        | 100       | 100       |
| Total  | 1000      | 1200      | 1400      | 1500      |

Construct a component bar diagram with the help of the above data.

**Q4)** The score (marks) of 10 students of a class in Computer Fundamental course is 68, 70, 60, 64, 67, 50, 54, 60, 80, 74. Find arithmetic mean, geometric mean, harmonic mean, median and mode of the given marks in the course. [Hints: AM=64.7, GM=64.12, HM=63.53, Me=65.5 and Mo=60]

**Q5)** The profits of 1000 departmental stores are given below. The median profit of the departmental stores is Tk. 413.11 lakhs. Estimate the missing values and the modal profits of the departmental stores.

| Profit (Tk. lakhs)            | 300-325 | 325-350 | 350-375 | 375-400 | 400-425 | 425-450 | 450-475 | 475-500 |
|-------------------------------|---------|---------|---------|---------|---------|---------|---------|---------|
| Number of departmental stores | 5       | 17      | 80      | $f_1$   | 326     | $f_2$   | 88      | 9       |

**Solution:** Calculation of missing frequencies

| Profits | Frequency ( $f_i$ )          | Cumulative frequency ( $F$ ) |
|---------|------------------------------|------------------------------|
| 300-325 | 5                            | 5                            |
| 325-350 | 17                           | 22                           |
| 350-375 | 80                           | 102                          |
| 375-400 | $f_1$                        | $102+f_1$                    |
| 400-425 | 326                          | $428+f_1$                    |
| 425-450 | $f_2$                        | $428+f_1+f_2$                |
| 450-475 | 88                           | $516+f_1+f_2$                |
| 475-500 | 9                            | $525+f_1+f_2$                |
|         | $N = 1000 = 525 + f_1 + f_2$ |                              |

Here,  $f_1 + f_2 = 1000 - 525 = 475$

Given, median,  $Me = 413.11$ , so median lies in the interval 400-425 since 413.11 lies in this class.



We know, Median,  $Me = L + \frac{\frac{N}{2} - F_c}{f_m} \times c$

Here,  $\frac{N}{2} = \frac{100}{2} = 500, L = 400, F_c = 102 + f_1, f_m = 326$  and  $c = 25$ .

$$\therefore Me = L + \frac{\frac{N}{2} - F_c}{f_m} \times c$$

$$\Rightarrow 413.11 = 400 + \frac{500 - (102 + f_1)}{326} \times 25$$

$$\Rightarrow f_1 = 227.05 \approx 227$$

Hence  $f_2 = 475 - 227 = 248$ . So the missing frequencies are 227 and 248.

#### Calculation of mode

| Profits | Frequency ( $f_i$ ) |
|---------|---------------------|
| 300-325 | 5                   |
| 325-350 | 17                  |
| 350-375 | 80                  |
| 375-400 | 227                 |
| 400-425 | 326                 |
| 425-450 | 248                 |
| 450-475 | 88                  |
| 475-500 | 9                   |

The modal class is 400-425, since it contains the highest frequency. The formula for finding mode is

$$Mo = L + \frac{\Delta_1}{\Delta_1 + \Delta_2} \times c$$

Here,  $L = 400, \Delta_1 = 326 - 227 = 99, \Delta_2 = 326 - 248 = 78$  and  $c = 25$ .

$$\therefore Mo = 400 + \frac{99}{99 + 78} \times 25 = 413.98$$

**Q6) The mean deviation and geometric mean of two numbers are 3 and  $\sqrt{7}$  respectively. Find the two numbers, standard deviation and variance, coefficient of variation, harmonic mean and median.**

[Hints: For two numbers,  $MD = SD = \frac{R}{2}$ ] [Answers: 7, 1; 3 & 9; 75%, 1.75; Me = 4]

**Q7) Lives of two models of refrigerators in a recent survey were found as follows.**

| Life (Number of years) | Model-A | Model-B |
|------------------------|---------|---------|
| 0-2                    | 5       | 2       |
| 2-4                    | 16      | 7       |
| 4-6                    | 13      | 12      |
| 6-8                    | 7       | 19      |
| 8-10                   | 5       | 9       |
| 10-12                  | 4       | 1       |

i) What is the average life of each model of these refrigerators?

- ii) Which of the two models shows more uniformity?  
 iii) If a person wants to buy a new refrigerator, which one will he prefer?

[i)  $\bar{x}_A = 5.12$  Years &  $\bar{x}_B = 6.16$  Years ii)  $CV_A = 55.47\%$  &  $CV_B = 36.53\%$ , the lower the value of CV, the more consistent (uniform) the data iii) Will Prefer B]

**Q8) A researcher wants to know the relationships between the performance score and year of service in the Garments sector of Bangladesh. The data collected by the researcher is given below:**

| Employee              | A | B | C | D | E |
|-----------------------|---|---|---|---|---|
| Year of service (x)   | 3 | 5 | 7 | 2 | 8 |
| Performance score (y) | 5 | 6 | 2 | 3 | 4 |

1. Draw scatter plot. Is there any relationship between performance score and year of service? If yes, comment on the type of relationship.
2. Calculate Karl Pearson's correlation coefficient and comment on it. (Use the formula of r,  $r = -0.18$ )

**Q9) Ages of 10 couples are given below:**

|                    |    |    |    |    |    |    |    |    |    |    |
|--------------------|----|----|----|----|----|----|----|----|----|----|
| Age of husband (x) | 22 | 23 | 23 | 24 | 26 | 27 | 27 | 28 | 30 | 30 |
| Age of wives (y)   | 18 | 20 | 21 | 20 | 21 | 22 | 23 | 24 | 25 | 26 |

1. Determine the regression equation of the ages of wives on the age of husband.
2. Determine the regression equation of the ages of husband on the age of wives.
3. If wife's age is 30, compute the probable age of husband.
4. If husband's age is 40, compute the probable age of wives.
5. Calculate the correlation coefficient between husband's age and wives age.

[Hints: 1) regression equation of y on x,  $\hat{y} = \hat{a} + \hat{\beta}x$ ,  $\hat{\beta} = \frac{\sum_{i=1}^n x_i y_i - n\bar{x}\bar{y}}{\sum_{i=1}^n x_i^2 - n\bar{x}^2}$  and  $\hat{a} = \bar{y} - \hat{\beta}\bar{x}$ , 2) regression equation of x on y,  $\hat{x} = \hat{a} + \hat{\beta}_1 y$ ,  $\hat{\beta}_1 = \frac{\sum_{i=1}^n x_i y_i - n\bar{x}\bar{y}}{\sum_{i=1}^n y_i^2 - n\bar{y}^2}$  and  $\hat{a} = \bar{x} - \hat{\beta}_1 \bar{y}$ , 3) Put  $x = 30$  and find the value of y. 4) Similar as 3., 5)  $r = \sqrt{\hat{\beta} \times \hat{\beta}_1}$ .]

**Q10) A manufacturer of fluorescent tubes claims that his tubes have a mean lifetime of 1950 burning hours. A random sample of 100 tubes is taken from a day's output and tested for burning life. It is found to have a mean burning lifetime of 1900 hours with a standard deviation of 150 hours. Can the claim of the manufacturer be accepted at 5% level of significance? Also find P-value.**

**Hints:** Here, nothing is said about population and population variance is not known. But sample size is large (>100) so Z-test should be used.

**Assumptions:**

- i) The sample is drawn from a normal population.



- ii) The sample is randomly selected.
- iii) The sample is independent.

Hypothesis:

$$H_0: \mu = 1950 \text{ vs } H_1: \mu \neq 1950$$

Test Statistics:

$$Z = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} \sim N(0,1) \text{ under } H_0.$$

Decision Rule: Reject  $H_0$  if  $|Z_{cal}| > Z_{\alpha/2}$

Calculation:

Here,  $\bar{X} = 1900, \mu_0 = 1950, s = 150$  and  $n = 100$

$$\therefore Z = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} = -3.33$$

Let,  $\alpha = 5\% = 0.05$ , so  $Z_{\alpha/2} = Z_{0.025} = 1.96$

Statistical Decision:

Since  $|Z_{cal}| = 3.33 > Z_{0.025} = 1.96$ , so we can reject the null hypothesis ( $H_0$ ) at 5% level of significance.

Conclusion: The claim of the manufacturer is not accepted, that means tubes have not a mean lifetime of 1950 burning hours.

**P-value:** From the table of area under the standard normal distribution, the value of

$P(Z < Z_{cal}) = P(Z < -3.33) = 0.0004$ . Since it is two tail test, the P-value is  $2 \times 0.0004 = 0.0008$ . Since P-value is far less than 0.01, so the value of z is highly significant.

**Q11)** A process that produces bottles of shampoo, when operating correctly, produces bottles whose weights on average 20 ounces. A random sample of nine bottles from a single production run yielded the following weights

21.4, 19.7, 19.7, 20.6, 20.8, 20.1, 19.7, 20.3, 20.9.

Assuming that the population distribution is normal, test the hypothesis that the process is operating correctly at 5% level of significance. Also calculate 95% confidence interval for population mean.

**Hints:** Since the sample size is small and population variance is unknown,  $t$  test will be used.

Assumptions:

- i) The sample is randomly selected.
- ii) The sample is drawn from a normal population.

Hypothesis:

$$H_0: \mu = 20 \text{ vs } H_1: \mu \neq 20$$

Test Statistics:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \sim t_{n-1} \text{ under } H_0.$$

Decision Rule: Reject  $H_0$  if  $|t_{cal}| > t_{n-1, \alpha/2}$

Calculation:

Here,  $\bar{X} = 20.36$  (calculate mean of 9 observation),  $\mu_0 = 20$ ,  $s = 0.61$  and  $n = 9$

$$\therefore t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = 1.77$$

Let,  $\alpha = 5\% = 0.05$  and  $df = n - 1 = 9 - 1 = 8$

So, from t table,  $t_{n-1, \alpha/2} = t_{14, 0.025} = 2.316$

Statistical Decision:

Since  $|t_{cal}| = 1.77 < t_{14, 0.025} = 2.316$ , so we fail to reject the null hypothesis ( $H_0$ ) at 5% level of significance.

Conclusion: The data support that the process is operating correctly.

95% confidence interval (CI) for population mean  $\mu$  is given by

$$\bar{X} \pm t_{n-1, \alpha/2} \cdot s/\sqrt{n} = (19.89, 20.83)$$

**Q12)** Consider an experiment of tossing two balanced dice and record the number of dots on the top faces. Let  $x$  denotes the number of top face of the first die and  $y$  be the number of top face of the second die. List the sample face of the experiment. Find the probabilities of the following events:

- $x + y = 7$
- $x \geq 5$  and  $y \leq 5$
- $x + y = \text{odd}$
- $x + y > 10$
- $x = 2y$
- $x \geq 5$  or  $y \leq 2$

Hints:

| X \ Y | 1     | 2     | 3     | 4     | 5     | 6     |
|-------|-------|-------|-------|-------|-------|-------|
| 1     | (1,1) | (1,2) | (1,3) | (1,4) | (1,5) | (1,6) |
| 2     | (2,1) | (2,2) | (2,3) | (2,4) | (2,5) | (2,6) |
| 3     | (3,1) | (3,2) | (3,3) | (3,4) | (3,5) | (3,6) |
| 4     | (4,1) | (4,2) | (4,3) | (4,4) | (4,5) | (4,6) |
| 5     | (5,1) | (5,2) | (5,3) | (5,4) | (5,5) | (5,6) |
| 6     | (6,1) | (6,2) | (6,3) | (6,4) | (6,5) | (6,6) |

Sample space,  $S = \{(x, y), x = 1, 2, 3, 4, 5, 6 \text{ and } y = 1, 2, 3, 4, 5, 6\}$

$\therefore n(s) = 36$

- Let the event,  $A = x + y = 7$ , so  $A = \{(1,6), (2,5), (3,4), (4,3), (5,2), (6,1)\}$

So,  $n(A) = 6$

$$P(A) = \frac{n(A)}{n(S)} = \frac{6}{36} = \frac{1}{6}$$

ii)  $P(B) = \frac{5}{18}$  iii)  $P(C) = \frac{1}{2}$  iv)  $P(D) = \frac{1}{18}$  v)  $P(E) = \frac{1}{12}$  vi)  $P(F) = \frac{5}{9}$

**Q13)** Two components  $a$  and  $b$  are connected in parallel in a simplified circuit. The circuit functions if either or both of the components are operational. It is known from the past evidence that probability of operating  $a$  is 0.99, that of operating  $b$  is 0.98 and that of operating both  $a$  and  $b$  together is 0.9702. Find the probability that the circuit is functioning.

**Hints:** Consider events,  $A$ : Component  $a$  operates.  $B$ : Component  $b$  is operates.  $P(A) = 0.99, P(B) = 0.98, P(A \cap B) = 0.9702$

The event,  $(A \cup B)$  describes the circuit is functioning. So,  $P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.9998$

So, there is a 99.98% chance to function the circuit.

**Q14)** A continuous random variable has the following probability density function  $f(x) = kx^2; 0 \leq x \leq 1$ .

- i) Find the value of  $k$ .
- ii) Find,  $P[0.2 \leq X \leq 0.5], P[X < 0.3], P[X > 0.75]$ .

**Hints:** i) Since the total probability is one. So  $\int_{-\infty}^{\infty} f(x)dx = 1$ ,  $\int_0^1 kx^2 dx = 1 \therefore k = 3$ .

ii)  $P[0.2 \leq X \leq 0.5] = \int_{0.2}^{0.5} 3x^2 dx = 0.117$ ,  $P[X < 0.3] = \int_0^{0.3} 3x^2 dx = 0.027$ ,  $P[X > 0.75] = \int_{0.75}^1 3x^2 dx = 0.578$ .

**Q15)** A bakery has the following probability function of daily demand for birthday cake.

|                                 |      |      |      |      |   |     |      |      |
|---------------------------------|------|------|------|------|---|-----|------|------|
| Number of cakes demanded $X: x$ | 1    | 2    | 3    | 4    | 5 | 6   | 7    | 8    |
| Probability $f(x)$              | 0.02 | 0.07 | 0.09 | 0.12 | k | 0.2 | 0.18 | 0.02 |

- i) Find the value of  $k$ ?
- ii) Find the expected number of birthday cakes demanded per day.

**Hints:** i) Since,  $\sum f(x) = 1$ , so  $0.02+0.07+0.09+0.12+k+0.2+0.18+0.02=1$ . So,  $k=0.3$

ii) Expected number of birthday cakes demanded is  $E(X) = \sum_{i=1}^8 x_i \cdot f(x_i) = x_1 f(x_1) + \dots + x_8 f(x_8)$   
 $= 5.03 \approx 5$

So, on average, the expected number of birthday cakes demanded per day is 5.

----- The End -----